

# Data Analytics Unit 3 — Complete Hinglish Summary + Mind Map Notes

## Unit 3: Exploratory Data Analysis & Visualization

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### 1. Exploratory Data Analysis (EDA)

#### Definition

EDA ka matlab hota hai data ko deeply analyze karna taaki:

- Patterns mile
- Trends samajh aaye
- Outliers detect ho
- Insights generate ho

Simple line:

“EDA = Data ko samajhne ki process.”

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#### EDA Goals

- Better understanding
  - Data cleaning support
  - Visualization
  - Decision making
  - Feature understanding
- 

#### EDA Mind Map

```
EDA
|
├── Data Understanding
├── Pattern Detection
├── Trend Analysis
├── Outlier Detection
├── Visualization
└── Business Insights
```

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## 2. Statistics & Probability for Data Analysis

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### Distribution

#### Meaning

Distribution batata hai ki data ka spread kaisa hai.

Example:

- Exam marks
- Heights
- Salaries

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### Types of Distribution

#### (A) Normal Distribution

Features:

- Bell shaped
- Symmetric
- Mean = Median = Mode

Examples:

- Height
- IQ
- Marks

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#### (B) Positive Skew Distribution

Features:

- Right side tail long
- Mean > Median > Mode

Example:

- Income distribution

## (C) Negative Skew Distribution

Features:

- Left side tail long
- Mean < Median < Mode

Example:

- Easy exam marks
- 

## Distribution Mind Map

```
Distribution
|
├─ Normal
├─ Positive Skew
└─ Negative Skew
```

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## 3. Mean

### Definition

Average value of data.

### Formula

Mean = Sum of observations / Number of observations

### Example

Data: 10,20,30,40

Mean:

$(10+20+30+40)/4$

$=100/4$

$=25$

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# Uses of Mean

- Average salary
  - Average marks
  - Business analytics
- 

## 4. Standard Deviation (SD)

### Definition

Data mean se kitna spread hai.

- Low SD → Data close hai
  - High SD → Data spread out hai
- 

### Numerical Example

Data: 2,4,4,4,5,5,7,9

Mean = 5

Squared differences:

$$(2-5)^2 = 9 \quad (4-5)^2 = 1 \quad (4-5)^2 = 1 \quad (4-5)^2 = 1 \quad (5-5)^2 = 0 \quad (5-5)^2 = 0 \quad (7-5)^2 = 4 \quad (9-5)^2 = 16$$

Sum = 32

Variance =  $32/8 = 4$

SD =  $\sqrt{4} = 2$

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## Applications

- Finance
  - Risk analysis
  - Market volatility
-

# SD Mind Map

Standard Deviation

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├ Low SD → Close Data

└ High SD → Spread Data

## 5. Median

### Definition

Sorted data ka middle value.

### Example

2,4,6,8,10

Median = 6

### Even Number Example

2,4,6,8

Median =  $(4+6)/2$

=  $10/2$

=5

### Example

2,4,6,8,10

Median = 6

## Why Important?

Outliers se affect nahi hota.

Example: 10,20,30,1000

Median better represent karega.

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## 6. Mode

### Definition

Most frequent value.

Example: 1,2,2,3

Mode = 2

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### Numerical Example

Data: 5,7,7,8,9,7,10

Most repeated value = 7

Mode = 7

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## Uses

- Customer preference
  - Most sold product
  - Common category
- 

## 7. Confidence Interval (CI)

### Definition

Estimated range jisme actual population value hone ke chances hote.

Example:  $170 \pm 2$

Range: 168 to 172

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## Numerical Example

Suppose:

Mean = 50 SD = 10 n = 25 z = 1.96

CI =  $50 \pm 1.96(10/\sqrt{25})$

=  $50 \pm 1.96(2)$

=  $50 \pm 3.92$

Confidence Interval: 46.08 to 53.92

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## Applications

- Medical surveys
  - Poll analysis
  - Market research
- 

## 8. Regression Analysis

### Definition

Variables ke relationship ko analyze karta.

Mostly prediction ke liye use hota.

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## Linear Regression

Equation:

$$y = mx + b$$

Where:

- y = prediction
  - m = slope
  - b = intercept
-

## Numerical Example

Equation:

$$y = 5x + 10$$

If  $x = 2$

Then:

$$y = 5(2) + 10$$

$$= 10 + 10$$

$$= 20$$

Predicted value = 20

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## Applications

- Sales forecasting
  - Demand prediction
  - Stock prediction
- 

## Regression Mind Map

```
Regression
|
├── Relationship Analysis
├── Prediction
├── Applications
│   ├── Sales Forecasting
│   ├── Stock Prediction
│   └── Demand Forecasting
```

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## 9. Hypothesis Testing

### Definition

Check karta hai claim true hai ya nahi.



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## Types

### Null Hypothesis ( $H_0$ )

Default statement.

### Alternative Hypothesis ( $H_1$ )

Opposite statement.

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## p-value

If:

$$p < 0.05$$

Then: Reject null hypothesis.

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## Example

Suppose p-value = 0.03

Since:  $0.03 < 0.05$

Result: Reject null hypothesis.

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## Applications

- Product testing
  - Medical trials
  - A/B testing
- 

## 10. Random Variables

### Definition

Random experiment pe depend value.

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# Types

## Discrete Random Variable

Countable values.

Examples:

- Dice
  - Student count
- 

## Continuous Random Variable

Infinite values.

Examples:

- Height
  - Weight
  - Temperature
- 

# 11. Probability Distribution

## Definition

Probability ka mathematical representation.

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# Types

## Binomial Distribution

Fixed number of trials.

Example: Coin toss.

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## Numerical Example

3 coin tosses me exactly 2 heads ki probability:

$$n = 3 \quad k = 2 \quad p = 0.5$$

$$P(X=2)$$

$$= 3C2 \times (0.5)^2 \times (0.5)$$

$$= 3 \times 0.25 \times 0.5$$

$$= 0.375$$

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## Normal Distribution

Most important continuous distribution.

Used in:

- AI
  - ML
  - Statistics
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# 12. Bayes Theorem

## Definition

Conditional probability theorem.

Applications:

- Spam filtering
  - Fraud detection
  - Medical diagnosis
- 

## Numerical Example

Suppose:

$$P(A) = 0.01 \quad P(B | A) = 0.99 \quad P(B) = 0.05$$

Using Bayes theorem:

$$P(A | B) = (0.99 \times 0.01) / 0.05$$

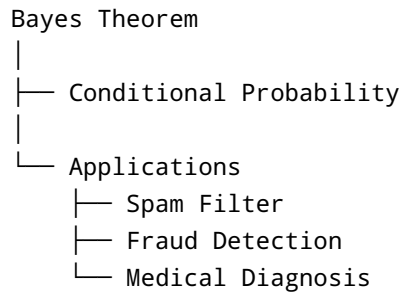
$$= 0.0099 / 0.05$$

$$= 0.198$$

Answer = 19.8%

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## Bayes Mind Map



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## 13. Data Visualization

### Definition

Data ko charts & graphs me represent karna.

Purpose:

- Fast understanding
  - Better insights
  - Trend analysis
- 

### Common Charts

| Chart        | Use          |
|--------------|--------------|
| Bar Chart    | Comparison   |
| Line Chart   | Trends       |
| Pie Chart    | Percentage   |
| Scatter Plot | Relationship |
| Histogram    | Distribution |
| Heatmap      | Density      |

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# Visualization Mind Map

```
Visualization
|
├─ Bar Chart
├─ Line Chart
├─ Pie Chart
├─ Scatter Plot
├─ Histogram
└─ Heatmap
```

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## 14. Power BI

### Definition

Microsoft ka Business Intelligence tool.

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### Features

- Dashboards
  - KPI monitoring
  - Real-time reports
  - Drag-drop interface
  - Data connectivity
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### Components

- Power BI Desktop
  - Power BI Service
  - Power Query
  - DAX
- 

### Applications

- Sales dashboards
  - Business reporting
  - KPI tracking
-

# 15. Tableau

## Definition

Advanced data visualization tool.

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## Features

- Beautiful charts
  - Fast analytics
  - Interactive dashboards
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## Components

- Tableau Desktop
  - Tableau Server
  - Tableau Prep
- 

## Applications

- Business analytics
  - Market analysis
  - Customer analysis
- 

## Power BI vs Tableau

| Feature       | Power BI  | Tableau            |
|---------------|-----------|--------------------|
| Cost          | Cheaper   | Expensive          |
| Visualization | Good      | Excellent          |
| Ease          | Easy      | Moderate           |
| Best For      | Reporting | Advanced Analytics |

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## 16. Dynamic Dashboards

### Features

- Filters
  - Slicers
  - Drill-down
  - Real-time updates
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### Dashboard Creation Steps

1. Data Collection
  2. Data Cleaning
  3. Data Modeling
  4. Visualization Selection
  5. Add Interactivity
  6. Publish Dashboard
- 

## 17. KPI (Key Performance Indicator)

### Meaning

Business performance metrics.

Examples:

- Revenue
  - Profit
  - Conversion rate
  - Customer retention
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## 18. Interactive Case Studies

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### Retail Store Analysis

Visualizations:

- Sales charts
- Region heatmaps

Insight: Weak sales regions identified.

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## COVID Dashboard

Visualizations:

- Maps
- Trend charts

Insight: High-risk regions identified.

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## E-commerce Dashboard

Problem: Cart abandonment.

Finding: Mobile users checkout pe leave kar rahe the.

Solution: Mobile UI improve ki gayi.

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## Banking Fraud Detection

Visualizations:

- Fraud maps
- Heatmaps

Result: Suspicious activities quickly detect hui.

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## Best Visualization Practices

- Simple charts use karo
  - Correct chart choose karo
  - Too many colors avoid karo
  - Important KPIs highlight karo
  - Interactivity add karo
- 

## Final Ultra Quick Revision

```
Unit 3
|
|— EDA
|   |— Trends
```



- | | | Patterns
- | | | Outliers
- | |
- | | Statistics
- | | | Mean
- | | | Median
- | | | Mode
- | | | SD
- | | | CI
- | |
- | | Probability
- | | | Random Variables
- | | | Probability Distribution
- | | | Bayes Theorem
- | |
- | | Regression
- | | Hypothesis Testing
- | |
- | | Visualization
- | | | Charts
- | | | Dashboards
- | | | KPIs
- | |
- | | Power BI
- | | Tableau